

Last login: Thu Jun 18 19:47:37 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# SET ROLE alice;
SET

tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT COUNT(*)
FROM (

```
SELECT l_orderkey, l_linenumbr
FROM lineitem
WHERE l_orderkey > 2000000
) l_alias;
```

WARNING: Log 1: Planner Input:

```
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumbr")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumbr"
"l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate"
"l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationk
ey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:joinTree
{FROMEXPR
```

```

        :fromlist (
        {RANGETBLREF
        :rtindex 1
        }
        )
        :quals
        {OPEXPR
        :opno 674
        :opfuncid 297
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
        {VAR
        :varno 1
        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
        }
        {FUNCEXPR
        :funcid 316
        :funcresulttype 701
        :funcset false
        :funcvariadic false
        :funcformat 2
        :funccollid 0
        :inputcollid 0
        :args (
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -24 3 0 0 0 0 0 0 ]
        }
        )
        :location -1
        }
        )
        :location -1
        }
        }
        :mergeActionList <>
        :mergeTargetRelation 0
        :mergeJoinCondition <>
        :targetlist (
        {TARGETENTRY
        :expr
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
        }
        )
        :override 0
        :onConflict <>
        :returningOldAlias <>
        :returningNewAlias <>
        :returningList <>
        :groupClause <>
        :groupDistinct false
        :groupingSets <>
        :havingQual <>
        :windowClause <>
        :distinctClause <>
        :sortClause <>
        :limitOffset <>
        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
        }
        :location -1
        }
        )
        )
        :rteperminfos (
        {RTEPERMISSIONINFO
        :relid 16566
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 11)
        :insertedCols (b)
        :updatedCols (b)
        }
        )
        :jointrue
        {FROMEXPR
        :fromlist (
        {RANGETBLREF
        :rtindex 1
        }
        )
        :quals
        {OPEXPR
        :opno 521
        :opfuncid 147
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
        {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 123
        }
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0

```

```

        :constlen 4
        :constbyval true
        :constisnull false
        :location 136
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
}
:location 134
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 70
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 82
}
:resno 2
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCt true
:securityQuals <>
}
)
:rteperminfos <>
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>

```

```

:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTF false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
)

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval tr
ue :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :ha
s_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :left_relds (b 1) :right_relds (b 1) :orclause <> :rinfo_serial 1 :
eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1
:left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval t

```

```

rue :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 64 -113 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :h
as_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1
:eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1
:left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: {PATH
:pathtype 339
:parent_relids (b 1)
:pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs ()
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 8196
:disabled_nodes 0
:startup_cost 0
:total_cost 343
:pathkeys <>
}

WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumber")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_qua
ntity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
))
})
}
:rtepermissions (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{FROMEXPR
:fromlist (
{RANGETBENTRY
:rteindex 2
}
)
:quals (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
}
)
}
)
}
}

```

```

:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
}
:location 134
}
}
}
}
:quals <>
}
:mergeActionList <>
:mergeTargetAction 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 43
}
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
)
WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_rels (b) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=0 in_safe_list=1 + reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_rels (b) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: (AGGPATH
:path.pathtype 365
:parent_rels (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 93400.51604887121
:path.total_cost 93400.52604887121
:path.pathkeys <>
:subpath
{GATHERPATH
:path.pathtype 368
:parent_rels (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 2
:path.disabled_nodes 0
:path.startup_cost 93400.25919918844
:path.total_cost 93400.46919918843
:path.pathkeys <>
:subpath
{AGGPATH
:path.pathtype 365
:parent_rels (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe true

```

```

:path.parallel_workers 2
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 92400.25919918844
:path.total_cost 92400.26919918843
:path.pathkeys <>
:subpath
{INDEXPATH
:path.pathtype 342
:parent_relids (b 2)
:path.target
{PATHTARGET
:exprs <>
:sortgrouprefs <>
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:path.parallel_aware true
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 830673
:path.disabled_nodes 0
:path.startup_cost 0.4325
:path.total_cost 90323.57669918844
:path.pathkeys <>
:indexinfo
{INDEXOPTINFO
:indexoid 17007
:reltablespace 0
:pages 16457
:tuples 6001215
:tree_height 2
:ncolumns 2
:nkeycolumns 2
:indexkeys ( 1 4)
:indexcollations ( 0 0)
:opfamily ( 1976 1976)
:opcintype ( 23 23)
:sortopfamily ( 1976 1976)
:reverse_sort ( false false)
:nulls_first ( false false)
:canreturn ( true true)
:relam 403
:indpred <>
:indexlist (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:indrestrictinfo (
{RESTRICTINFO
:clause
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 0
:clause_relids (b)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b)
:right_relids (b)
:orclause <>
:rinfo_serial 1
:eval_cost.startup 0
:eval_cost.per_tuple 0
:norm_selec 0.5
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
{RESTRICTINFO
:clause
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR

```

```

        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location 123
    }
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 136
    :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
    }
    :location 134
    }
    :is_pushed_down true
    :can_join false
    :pseudoconstant false
    :has_clone false
    :is_clone false
    :leakproof true
    :has_volatile 0
    :security_level 1
    :num_base_rels 1
    :clause_relids (b 2)
    :required_relids (b 2)
    :incompatible_relids (b)
    :outer_relids (b)
    :left_relids (b 2)
    :right_relids (b)
    :orclause <>
    :rinfo_serial 2
    :eval_cost.startup 0
    :eval_cost.per_tuple 0.0025
    :norm_selec 0.6644041354651555
    :outer_selec -1
    :mergeopfamilies <>
    :left_em <>
    :right_em <>
    :outer_is_left false
    :hashjoinoperator 0
    :left_bucketsize -1
    :right_bucketsize -1
    :left_mcvfreq -1
    :right_mcvfreq -1
    :left_hasheqoperator 0
    :right_hasheqoperator 0
    }
    )
    :predOK false
    :unique false
    :nullsnotdistinct false
    :immediate true
    :hypothetical false
    :amcanorderbyop false
    :amoptionalkey true
    :amsearcharray true
    :amsearchnulls true
    :amhasgettupple true
    :amhasgetbitmap true
    :amcanparallel true
    :amcanmarkpos true
    }
    :indexclauses (
    {INDEXCLAUSE
    :rinfo
    {RESTRICTINFO
    :clause
    {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location 123
    }
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 136
    :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
    }
    :location 134
    }
    :is_pushed_down true
    :can_join false
    :pseudoconstant false
    :has_clone false
    :is_clone false
    :leakproof true
    :has_volatile 0
    :security_level 1
    :num_base_rels 1
    :clause_relids (b 2)
    :required_relids (b 2)
    :incompatible_relids (b)
    :outer_relids (b)
    :left_relids (b 2)
    :right_relids (b)
    :orclause <>
    :rinfo_serial 2
    :eval_cost.startup 0
    :eval_cost.per_tuple 0.0025
    :norm_selec 0.6644041354651555
    :outer_selec -1
    :mergeopfamilies <>
    :left_em <>
    :right_em <>
    :outer_is_left false
    :hashjoinoperator 0
    :left_bucketsize -1
    :right_bucketsize -1
    :left_mcvfreq -1
    :right_mcvfreq -1
    :left_hasheqoperator 0
    :right_hasheqoperator 0
    }
    )
    :indexquals (
    {RESTRICTINFO
    :clause

```

```

{OPEXPR
:opno 521
:opfncid 147
:oprresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
  {CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relds (b 2)
:required_relds (b 2)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b 2)
:right_relds (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_fm <>
:right_fm <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
)
:lossy false
:indexcol 0
:indexcols <>
}
)
:indexorderbys <>
:indexorderbycols <>
:indexscandir 1
:indextotalcost 73644.6725
:indexselectivity 0.6644041354651555
}
:aggstrategy 0
:aggsplit 6
:numGroups 1
:transitionSpace 0
:groupClause <>
:qual <>
}
:single_copy false
:num_workers 2
}
:aggstrategy 0
:aggsplit 9
:numGroups 1
:transitionSpace 0
:groupClause <>
:qual <>
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded true
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 93400.51604887121
:plan.total_cost 93400.52604887121
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2003
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtransype 20
:aggargtypes <>
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
}
}
:resno 1

```

```

        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 9
:aggn 0
:aggtransno 0
:location 43
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{GATHER
:plan.disabled_nodes 0
:plan.startup_cost 93400.25919918844
:plan.total_cost 93400.46919918843
:plan.plan_rows 2
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 92400.25919918844
:plan.total_cost 92400.26919918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 6
:aggn 0
:aggtransno 0
:location 43
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{INDEXONLYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 90323.57669918844
:scan.plan.plan_rows 830673
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 1
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -3

```

```

        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 4
        :location -1
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paratype 16
:paratypmod -1
:paramcollid 0
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanrelid 2
:indexid 17007
:indexqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
)
:recheckqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
)
:indexorderby <>
:indexlist (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2

```

```

        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
    :indexorderdir 1
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0 1)
    :plan.allParam (b 0 1)
    :aggstrategy 0
    :aggssplit 6
    :numCols 0
    :grpColldx ()
    :grpOperators ()
    :grpCollations ()
    :numGroups 1
    :transitionSpace 0
    :aggParams (b)
    :groupingSets <>
    :chain <>
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0)
    :plan.allParam (b 0)
    :num_workers 2
    :rescan_param 1
    :single_copy false
    :invisible false
    :initParam (b 0)
    }
    :plan.righttree <>
    :plan.initPlan (
    {SUBPLAN
        :subLinkType 0
        :testexpr <>
        :paramIds <>
        :plan_id 1
        :plan_name InitPlan\ 1
        :firstCollType 2278
        :firstCollTypmod -1
        :firstColCollation 0
        :useHashTable false
        :unknownEqFalse true
        :parallel_safe true
        :setParam (i 0)
        :parParam <>
        :args <>
        :startup_cost 0
        :per_call_cost 0.041849682772083945
    }
    )
    :plan.extParam (b)
    :plan.allParam (b 0)
    :aggstrategy 0
    :aggssplit 9
    :numCols 0
    :grpColldx ()
    :grpOperators ()
    :grpCollations ()
    :numGroups 1
    :transitionSpace 0
    :aggParams (b)
    :groupingSets <>
    :chain <>
    }
    :partPruneInfos <>
    :rttable (
    {RANGETBLENTTRY
        :alias
        {ALIAS
            :aliasname l_alias
            :colnames <>
        }
        :eref
        {ALIAS
            :aliasname l_alias
            :colnames ("l_orderkey" "l_linenumber")
        }
        :rtekind 1
        :subquery <>
        :security_barrier false
        :relid 0
        :inh false
        :relkind <>
        :rellockmode 0
        :perminfoindex 0
        :lateral false
        :inFromCl true
        :securityQuals <>
    }
    {RANGETBLENTTRY
        :alias <>
        :eref
        {ALIAS
            :aliasname lineitem
            :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesstatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
        }
        :rtekind 0
        :relid 16566
        :inh false
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
    }
    {RANGETBLENTTRY
        :alias
        {ALIAS
            :aliasname s
            :colnames <>
        }
        :eref
        {ALIAS
            :aliasname s
            :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
        }
        :rtekind 0
        :relid 16541
        :inh false
        :relkind r
        :rellockmode 1
        :perminfoindex 2
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
    }
    )
    :unprunableRelids (b 2 3)
    :permInfos (
    {RTEPERMISSIONINFO
        :relid 16566
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 11)
    }
    )

```

```

:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 343
:scan.plan.plan_rows 8196
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16566 16541)
:invalItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: (PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 92400.25919918844
:plan.total_cost 92400.26919918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggresorted false
:agglevelsup 0
:aggssplit 6
:aggno 0
:aggtransno 0
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{INDEXONLYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 90323.57669918844
:scan.plan.plan_rows 830673
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false

```

```

:scan.plan.plan_node_id 3
:scan.plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno -3
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno -3
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 4
      :location -1
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:scan.plan.qual (
  {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 16
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanreldid 2
:indexid 17007
:indexqual (
  {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
      }
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
:recheckqual (
  {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
      }
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
:indexorderby <>
:indexextlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
    }
  }
)

```

```

:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 4
:varatype 23
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggssplit 6
:numcols 0
:grpColldx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumber")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 2 3)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
}
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN

```

```

:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 343
:scan.plan.plan_rows 8196
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
  {OEXPR
    :opno 674
    :opfuncid 297
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 6
        :location -1
      }
      {CONST
        :consttype 701
        :consttypmod -1
        :constcollid 0
        :constlen 8
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
      }
    )
    :location -1
  }
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL:      {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 0
:planTree
  {AGG
    :plan.disabled_nodes 0
    :plan.startup_cost 92400.25919918844
    :plan.total_cost 92400.26919918843
    :plan.plan_rows 1
    :plan.plan_width 8
    :plan.parallel_aware false
    :plan.parallel_safe true
    :plan.async_capable false
    :plan.plan_node_id 2
    :plan.targetlist (
      {TARGETENTRY
        :expr
          {AGGREF
            :aggfnoid 2803
            :aggtype 20
            :aggcollid 0
            :inputcollid 0
            :aggtranstype 20
            :aggargtypes <>
            :aggdirectargs <>
            :args <>
            :aggorder <>
            :aggdistinct <>
            :aggfilter <>
            :aggstar true
            :aggvariadic false
            :aggkind n
            :aggpresorted false
            :agglevelsup 0
            :aggssplit 6
            :aggno 0
            :aggtransno 0
            :location -1
          }
        :resno 1
        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
    :plan.qual <>
    :plan.lefttree
      {INDEXONLYSCAN
        :scan.plan.disabled_nodes 0
        :scan.plan.startup_cost 0.4325
        :scan.plan.total_cost 90323.57669918844
        :scan.plan.plan_rows 830673
        :scan.plan.plan_width 0
        :scan.plan.parallel_aware true
        :scan.plan.parallel_safe true
        :scan.plan.async_capable false
        :scan.plan.plan_node_id 3
        :scan.plan.targetlist (
          {TARGETENTRY
            :expr
              {VAR
                :varno -3
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 2
                :varattnosyn 1
                :location -1
              }
            }
          }
        )
      }
    )
  }
}

```

```

:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0 1)
:scan.plan.allParam (b 0 1)
:scan.scanrelid 2
:indexid 17007
:indexqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location -1
}
)
:recheckqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location -1
}
)
:indexorderby <>
:indextlist (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 4

```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 4
        :location -1
    }
    :resno 2
    :resname <>
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
}
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggsplit 6
:numCols 0
:grpColldx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumber")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 2 3)
:permInfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
}
:resultRelations <>
:appendRelations <>
:subplans (
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 343
:scan.plan.plan_rows 8196
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist <>
:scan.plan.qual (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0

```

```

:inputcollid 0
:args (
  (VAR
    :varno 3
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 6
    :location -1
  )
  (CONST
    :consttype 701
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
  )
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 3
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalidItems <>
:paramExecTypes (o 16 0)
:utilityStat <>
:stmt_location -1
:stmt_len -1
}

```

QUERY PLAN

```

Finalize Aggregate  (cost=93400.52..93400.53 rows=1 width=8) (actual time=227.680..229.175 rows=1.00 loops=1)
  Output: count(*)
  Buffers: shared hit=10991
  InitPlan 1
    -> Seq Scan on public.supplier s  (cost=0.00..343.00 rows=8196 width=0) (actual time=0.031..0.031 rows=1.00 loops=1)
        Filter: (s.s_acctbal > '1000'::double precision)
        Buffers: shared hit=2
    -> Gather  (cost=93400.26..93400.47 rows=2 width=8) (actual time=227.555..229.168 rows=3.00 loops=1)
        Output: (PARTIAL count(*))
        Workers Planned: 2
        Workers Launched: 2
        Buffers: shared hit=10991
        -> Partial Aggregate  (cost=92400.26..92400.27 rows=1 width=8) (actual time=221.215..221.215 rows=1.00 loops=3)
            Output: PARTIAL count(*)
            Buffers: shared hit=10989
            Worker 0:  actual time=217.224..217.224 rows=1.00 loops=1
                Buffers: shared hit=3633
            Worker 1:  actual time=219.301..219.301 rows=1.00 loops=1
                Buffers: shared hit=3654
            -> Parallel Index Only Scan using l_pk on public.lineitem  (cost=0.43..90323.58 rows=830673 width=0) (actual time=0.187..0.168.661 rows=1333572.67 loops=3)
                Output: lineitem.l_orderkey, lineitem.l_linenumber
                Index Cond: (lineitem.l_orderkey > 2000000)
                Filter: (InitPlan 1).col1
                Heap Fetches: 0
                Index Searches: 1
                Buffers: shared hit=10989
                Worker 0:  actual time=0.081..0.165.250 rows=1323456.00 loops=1
                    Buffers: shared hit=3633
                Worker 1:  actual time=0.093..0.166.771 rows=1331055.00 loops=1
                    Buffers: shared hit=3654

Planning:
  Buffers: shared hit=121
  Planning Time: 11.264 ms
  Execution Time: 233.162 ms
(34 rows)

tpch=>

```